Kent Neureiter

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Education

Lehigh University Bethlehem, PA

B.S. in Computer Engineering, GPA: 3.71/4.0, IBE Honors Student, Dean's List All Semesters Aug. 2024 – May 2028

- **Relevant Courses:** Data Structures & Algorithms, Systems Software, Computer Architecture, Digital Logic & Design, FPGA Lab, Signals & Systems, Software Engineering, Intro to Machine Learning, Discrete Structures, Object-Oriented Programming, Linear Algebra, Calculus I–III

Research Experience

Lehigh Autonomous and Intelligent Robotics (AIR) Lab Bethlehem, PA

Undergraduate Research Assistant under Prof. David Saldaña *June 2026 – Present*

- Developed a custom MuJoCo simulator for autonomous navigation control of a drone swarm, tuning differential equations for robotic control and sim-to-real transfer, and testing object detection and course correction during autonomous flight
- Profiled perception code of an autonomous drone stack to identify bottlenecks consuming 86% of total compute, then translated and cross-compiled computation-heavy functions to C++
- Cut total inference latency from 109ms to 57ms (47% improvement), with the largest gains from integrating a precomputed lookup table, eliminating redundant per-pixel floating-point computation in the video processing pipeline
- Integrated the optimized module into the existing stack, enabling faster autonomous detection and task execution across a multi-drone fleet
- Debugged embedded hardware from ESC calibration testing to checking PWM signals on an oscilloscope

Lehigh BINA Lab Bethlehem, PA

Undergraduate Research Assistant under Prof. Maryam Rahnemoonfar *May 2026 – June 2026*

- Contributing to 3D-LifeSeg-VQA, the first multi-task 3D point cloud dataset for post-disaster damage assessment, integrating instance segmentation, VQA annotations, reasoning segmentation, and scene captioning
- Designed and documented a VQA annotation framework covering 11 question types (counting, spatial reasoning, accessibility, proportional analysis) across scene and building levels, with automated generation pipelines for structured question-answer pairs
- Annotated UAV-derived 3D point cloud scenes for the dataset alongside a fellow undergraduate researcher
- Surveyed 3D VLM frameworks (3D-R1, CLEVR3D, LL3DA, LLaVA-3D) and presented comparative analysis of dataset construction methodologies to the research group, identifying applicable techniques for outdoor disaster scene annotation

Experience

Lehigh Mars Rover Team (University Rover Challenge) Bethlehem, PA

Software Team Co-Lead *Oct. 2025 – Present*

- Built drivetrain control software (dual-joystick input, per-wheel alignment) in C++/Python, deployed across distributed Raspberry Pi nodes over CAN bus and containerized with Docker for real-time rover control
- Wrote a ROS 2 C++ node that drives the rover to GPS waypoints, computing distance and bearing between coordinates and adjusting speed/heading with proportional control
- Collaborated across software, electrical, and mechanical sub-teams to integrate control software with hardware components during field testing and iteration

Astris Advisory Japan KK Tokyo, Japan

Summer Intern, Software Engineering and Research *June 2025 – Aug. 2025*

- Designed and demoed liking and commenting features end to end for a client-facing investment article page
- Added PostgreSQL join tables with composite keys, implemented 4 Java REST endpoints, and built Vue/TypeScript UI components
- Optimized feed loading by aggregating like counts in a single SQL join/count query, eliminating per-message database calls
- Wrote 30+ unit and UI tests (JUnit, Vitest) and delivered features through code-reviewed pull requests with a mentor

- Shipped a client-facing search and filtering (sector, date, keyword) feature for the firm's investment research platform
- Developed Python scripts integrating Refinitiv's API with pandas to scrape, clean, and analyze large-scale financial time series data for 1,000+ TOPIX-listed companies
- Automated extraction of industry-specific metrics (e.g., Price-to-Book ratios), enabling benchmarking across firms and saving several hours of manual research per project

Projects

Resonance News | *FastAPI, Next.js, Supabase* | resonance-news.com Mar. 2026 – Present

- Full-stack AI news platform delivering personalized daily briefings to 50+ active users
- Converts free-text onboarding responses into structured interest vectors, bootstrapping recommendations with no click history
- Designed an information retrieval pipeline that ingests 500+ articles/day from RSS feeds and the Guardian's API, embeds them with OpenAI's model, and applies Leiden graph clustering to group related coverage and deduplicate similar topics across outlets
- Implemented cosine similarity ranking against user interest vectors, scoring both topical focus and cross-topic relevance to keep briefings varied

RaiseIt: Hand Raising Detection | *YOLOv8, ONNX Runtime, TensorRT, Jetson* Feb. 2025 – May 2026

- Computer vision system to detect raised hands in large lecture halls using YOLOv8 with ByteTrack for multi-person tracking
- Deployed on a Jetson Orin Nano via ONNX Runtime with a systemd service for headless autostart; porting inference to TensorRT
- Fine-tuned the model on 300+ annotated images and deployed across 5 classrooms on campus

Digital Design & FPGA | *SystemVerilog, Vivado* | github.com/kentneureiter/fpga_labs 2025 – Present

- Designed two interacting finite state machines (theater lighting controller) from a requirements document, deriving state encodings and minimizing next-state logic with Karnaugh maps before implementing and verifying in SystemVerilog
- Built combinational and sequential designs in Verilog including a carry-lookahead adder, BCD to seven-segment decoder, and flip-flop and counter library, each with its own testbench
- Synthesized and deployed to a Basys-3 FPGA through Vivado, writing pin constraints and validating behavior on hardware

Park A Lot | *AWS Serverless* | AWS Hackathon, 3rd Place Oct. 2025

- Built a serverless real-time campus parking availability app in 36 hours on a 2-person team
- Integrated Amazon Location Service geofencing to confirm a driver is inside a lot before accepting a status update, so only on-site users can change availability
- Exposed lot status through a REST API on API Gateway backed by DynamoDB, keeping availability synced live across devices

Technical Skills

Languages: C++, Python, Java, SystemVerilog/Verilog, SQL, JavaScript/TypeScript, Bash
Robotics & ML: ROS 2, MuJoCo, YOLOv8, ONNX Runtime, TensorRT, OpenAI Embeddings
Web & Cloud: FastAPI, React/Next.js, Vue, PostgreSQL, Supabase, AWS (Lambda, API Gateway, DynamoDB, Amplify)
Tools: Git, Linux, Docker, CI/CD, Maven, JUnit, Vitest, Vivado, Claude Code
Hardware: Jetson Orin Nano, Raspberry Pi, Basys-3 FPGA, CAN bus, ESP-NOW, Oscilloscope
Spoken Languages: English (native), Japanese (proficient)